CHALLENGE SUBMISSION REPORT

Scalable, Safeguarded DOM Optimization

Business value, mathematical model, implementation, scaling, and hardware evidence

Andrei Pomorov

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Portal summary

This submission delivers a runnable distributed order management solver that assigns whole order groups to eligible distribution-center/date options and optimizes integer fulfillment under cumulative inventory, dock capacity, diversion, and service-penalty rules. The production hierarchy combines load-atomic greedy construction, exact fixed-policy recourse, and adaptive exact large-neighborhood search (LNS). A bounded QUBO/QAOA path is included as a proposal mechanism behind deterministic repair, exact recourse, strict improvement, and independent validation.

The final evidence contains 516 aggregate experiment rows across 14 families. All 513 returned plans pass validation with zero recorded demand, integrality, inventory, and capacity residuals; three frozen-routing controls are correctly proven infeasible under 60-70% inventory reductions. On the common 100-group subset, greedy improves objective capture by 1.86 percentage points and case fill by 1.62 points over default in 0.58 seconds. All scalable methods reach 372 real groups; greedy and exact LNS reach 100,000 generated orders, while the hybrid reaches 20,000 with a 32-variable local QUBO. An 18-job, 8,192-shot IBM study shows that shallow QAOA retains substantially stronger raw feasible-subspace quality than the deeper circuit. The recommendation is to deploy the classical hierarchy in shadow mode and retain QAOA as a controlled, measurable local proposal path.

1. Business and technical framing

The planner must decide where and when an order should ship, then determine how many cases of each SKU can be fulfilled. A useful recommendation must balance service, unmet-demand penalties, and shipping cost while respecting rules that make naive order-by-order assignment unsafe:

- orders on one source load move together;
- alternatives must be eligible for every SKU and date;
- cumulative shipment through each checkpoint cannot exceed protected ATP;
- documented dock and enabled operational capacities cannot be exceeded;
- diversion must improve the protected default preview by the required threshold; and
- all fulfilled and unmet case quantities remain nonnegative integers.

The optimized objective is fulfilled merchandise value minus thresholded unmet-demand penalty minus shipping cost. To keep the submission privacy-safe, real evidence is reported as **objective capture** - objective divided by total requested merchandise value - and **case fill**, fulfilled cases divided by requested cases. Raw source values, commercial totals, and identifiers are excluded.

The decision is not “classical or quantum.” It is how to build a planner-grade system that is useful at every stage of the evidence. The implementation therefore keeps a trusted feasible incumbent and treats every more expensive method as an escalation.

2. Data understanding and preparation

The adapter consumes five canonical runtime files. It resolves the documented challenge columns, normalizes dates and identifiers, validates numeric domains, and stops before model construction if a required file is absent, empty, or unreadable.

Runtime input	Optimization role	Key checks
Order and line demand	Requested cases, values, penalties, loads, defaults	Positive demand, complete load/order keys, valid thresholds
Candidate shipping alternatives	Eligible DC/date choices, lead times, shipping	Lane/date validity, SKU coverage, diversion and calendar filters
Projected ATP	Time-phased DC/SKU availability	Complete keys, numeric quantities, protected cumulative checkpoints
Throughput observations	Optional operational-rate scenarios	Explicit enablement; no invented hard limit from an observation
Remaining dock capacity	Source-fact fixed capacity	Valid DC/date joins and nonnegative remaining capacity

Candidate generation removes invalid lanes, late or closed dates, missing DC/SKU coverage, and exhausted documented capacity before optimization. Orders sharing a load are collapsed into assignment groups with a common option set. The model also adds an explicit unassigned outcome; it does not create a fake candidate row.

The full supplied scope contains 750 modeled orders, 372 assignment groups, 20,869 order lines, and 2,182 unpruned candidate rows. Heuristic Pareto pruning can reduce the candidate universe, but remains opt-in because isolated dominance is not a global proof under shared resources. Candidate-count sensitivity shows that the important quality transition is from one to two retained candidates; larger caps through six do not change the common-subset frontier.

3. Baselines and evaluation contract

The challenge asks for default and greedy baselines before advanced optimization. Both are implemented as whole-group policies and measured under the same objective and validator as every other method.

Method	Mechanism	Purpose
Default routing	Retain the planned route and allocate feasible quantities	Business baseline
Greedy	Rank eligible group options, commit the best feasible increment, update resources	Fast algorithmic baseline
Polished greedy	Fix greedy routes and solve exact quantity/penalty recourse	Routine planning candidate
Exact LNS	Reopen bounded, conflict-linked groups in a local MILP	Quality escalation
Full MILP	Open all selected decisions and report incumbent, bound, and gap	Small-scope comparator
Hybrid sampler	Sample a bounded assignment QUBO; repair and solve exact recourse	Research path

Every returned plan is independently checked for one outcome per order, group cohesion, demand identity, integrality, candidate eligibility, cumulative ATP, enabled capacity, diversion improvement, output schema, and objective recomputation. A solver status is never accepted as sufficient evidence. The final grid has 513 returned plans and all 513 pass; maximum recorded numeric residuals are zero. The three non-plan rows are not software failures: the exact recourse model proves the frozen nominal routing infeasible after inventory is reduced by 60%, 65%, and 70%.

4. Mathematical formulation

For each eligible candidate $c \in C_o$, binary x_c selects its DC/date. Binary z_o selects the unassigned outcome. Integer f_{osc} and u_{os} represent fulfilled and unmet cases for order o, SKU s. The objective is

$$max\ J = \sum_{o,s,c} v_{os} f_{osc} - \sum_{o} P_o - \sum_{c} C_c x_c .$$

The core constraints are

$$\sum_{c \in C_o} x_c + z_o = 1 , \sum_{c \in C_o} f_{osc} + u_{os} = Q_{os} , 0 \le f_{osc} \le Q_{os} x_c .$$

Group-linking equalities make all members of a load choose the same outcome. Protected time-phased inventory is cumulative:

$$\sum_{o} \sum_{c : d(c) = d , \tau(c) \le t} f_{osc} \le A_{dst} \forall d , s , t .$$

Capacity rows combine fixed candidate use and case-dependent use. A non-default candidate must meet the documented service improvement over the protected default preview. Thresholded penalties, floors, caps, pallet picks, and loose-case picks are linearized exactly when enabled. The full derivation and assumptions are in docs/mathematical_formulation.md and docs/assumptions.md.

5. Scalable implementation

5.1 Classical hierarchy

Greedy construction compiles immutable line and candidate records once, caches group options, and updates compact residual inventory/capacity state as each group is committed. Fixed-policy recourse then prunes every unused candidate column and solves only the quantity and penalty problem for the selected policy.

Exact LNS avoids a dense all-pairs conflict graph. A sparse inverted index maps each resource to groups that use it. Neighborhoods are discovered lazily, frozen incumbent consumption is residualized, and the active assignment and quantity variables are solved jointly. Limits on groups, orders, fulfillment variables, time, and gap make the search predictable. The best valid incumbent is retained after timeout or failure.

5.2 Bounded QUBO and QAOA

The hybrid changes only the local assignment proposal. For retained choices m_g in each active group,

$$n_{QUBO} = \sum_{g} m_g \le n_{max} .$$

The QUBO includes plan value, one-hot penalties, and pairwise conflict surrogates. Samples are evaluated before recourse, one-hot violations are repaired, duplicate assignments are removed, and the best candidates receive exact fixed-assignment recourse. Only a strict, validator-passed improvement may replace the incumbent.

Gate-model QAOA starts each group in a weight-one Dicke/W state and uses a connected XY path mixer with $m_g - 1$ logical edges. Angles are optimized in the exact feasible subspace and reused across matched repetitions. Exact enumeration and uniform-feasible sampling define transparent controls.

QAOA is the optimization proposal method. IBM's Qiskit backend term names the processor or execution target; it is not a solver. The IBM processor receives only an independently generated 16-variable control.

6. Algorithm and result comparison

6.1 Common 100-group subset

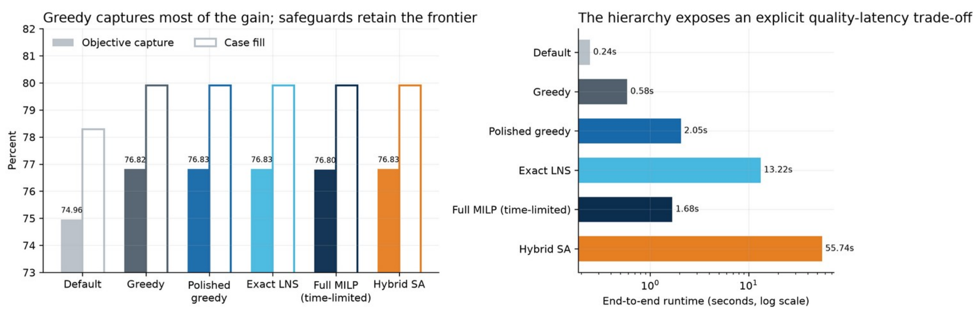
Method	Objective capture	Case fill	Reassigned	Runtime
Default	74.958%	78.288%	0	0.244 s
Greedy	76.817%	79.907%	13	0.583 s
Polished greedy	76.826%	79.907%	13	2.050 s
Exact LNS	76.826%	79.907%	13	13.225 s
Full MILP, time-limited	76.800%	79.908%	13	1.676 s
Hybrid simulated annealing	76.826%	79.907%	13	55.742 s

The two required baselines also report the objective's cost components in normalized, privacy-safe form:

Baseline	Penalty / requested value	Shipping / requested value
Default	0.4817%	0.6754%
Greedy	0.4232%	0.7060%

The executable notebook records the corresponding local totals; they are intentionally excluded from the public branch because the challenge privacy rule prohibits commercial cost disclosure.

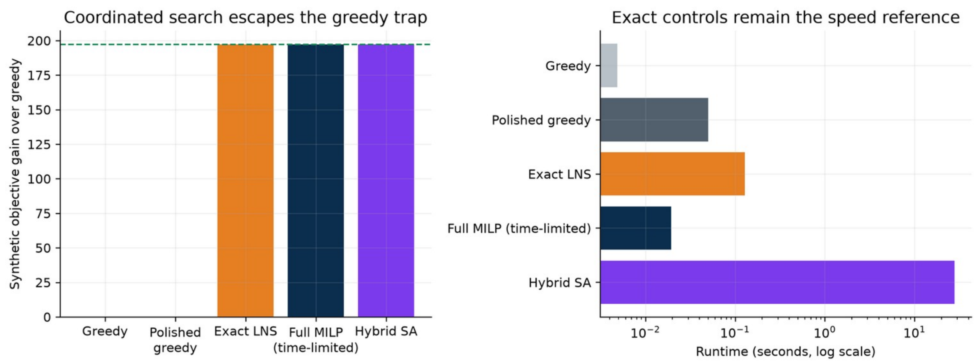
Greedy provides most of the nominal gain: +1.859 percentage points of objective capture and +1.619 points of case fill over default. Exact policy recourse adds 0.0085 capture points. Exact LNS makes only a negligible further move, and hybrid accepts no post-polish move on this nominal subset. The full MILP ends with a 0.0545% relative gap and lower incumbent, so it is correctly labeled a time-limited comparator.



Common solver comparison

6.2 Coordinated improvement control

The generated control creates a greedy trap that requires a coordinated assignment change. Greedy and polished greedy reach 21.49% fill. Exact LNS, full MILP, simulated annealing, local QAOA, and hardware-derived QAOA proposals reach 30.58% fill after recourse, a 197.2 synthetic objective-unit gain. Full MILP takes 0.019 seconds, exact LNS 0.129 seconds, and the simulated-annealing hybrid 27.873 seconds. The result proves that the decomposition can express coordinated moves and supplies a shared control for proposal-quality experiments.

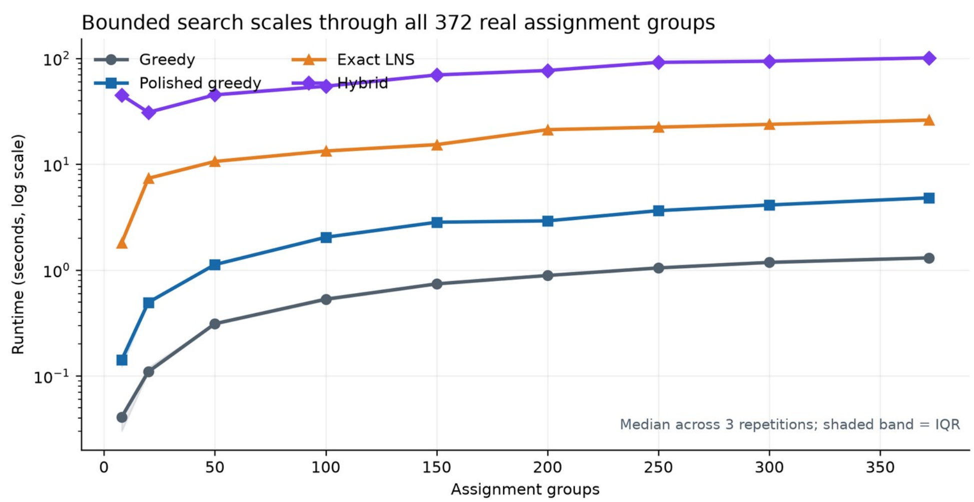


Coordinated control

7. Scaling, robustness, noise, and hardware

7.1 Real and generated scaling

All scalable methods reach the complete 372-group, 750-order real scope. Median runtime over three repetitions is 1.305 seconds for greedy, 4.816 for polished greedy, 26.143 for exact LNS, and 101.169 for hybrid simulated annealing. Exact LNS retains the highest full-scope capture, 82.122%, versus 82.115% for greedy.

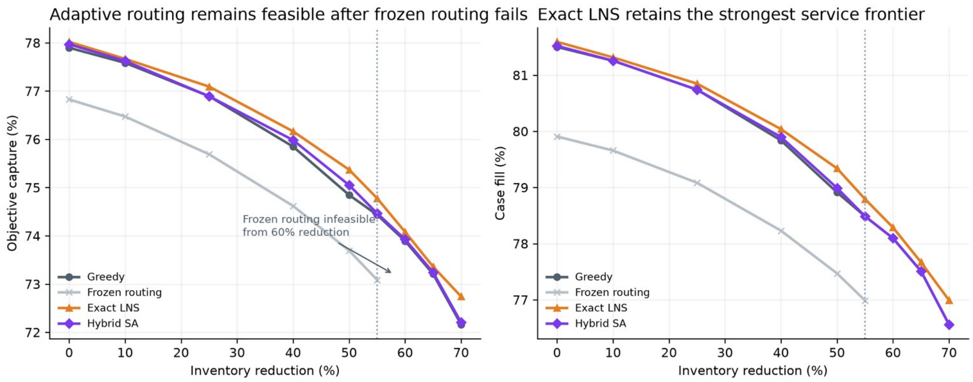


Repeated real scaling

Generated scaling reaches 100,000 orders for greedy (39.163-second median) and exact LNS (276.707 seconds). Hybrid scaling reaches 20,000 orders in 155.935 seconds while the local QUBO stays at 32 variables. The operational network can therefore grow without forcing the quantum-facing representation to grow with it.

7.2 Inventory robustness

Frozen routing remains feasible through a 55% inventory reduction and is then proven infeasible. Adaptive methods remain feasible through 70%. At 70%, objective capture is 72.162% for greedy, 72.204% for hybrid, and 72.741% for exact LNS; corresponding case fill is 76.560%, 76.560%, and 76.992%. This is the clearest business-facing advantage of adaptive coordination: it keeps producing valid routes after quantity-only recourse on the old policy breaks down.



Inventory robustness

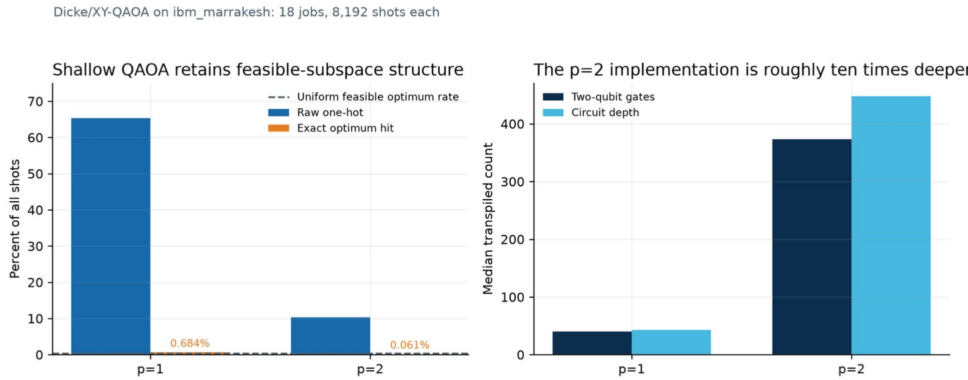
7.3 Noise and sampler controls

Exact feasible sampling, simulated annealing, and local constraint-preserving QAOA are one-hot by construction on the control; a random binary sampler is one-hot for only 0.352% of shots. Simulated annealing hits the exact QUBO optimum on 48.44% of shots and local QAOA on 1.21%, against a 0.3906% uniform-feasible rate. All recover the same final move after recourse, demonstrating why raw sampling and downstream plan quality must be reported separately.

QUBO coefficient noise degrades raw structure after roughly 10% relative perturbation; at 20% the median one-hot rate is about 29.4%. An independent readout-bit-flip proxy reduces one-hot feasibility to about 19.7% at a 10% flip probability. Every safeguarded row still recovers the control improvement. These are sensitivity studies, not a full physical QPU noise model.

7.4 IBM hardware and runtime correction

The final hardware matrix contains 18 successful jobs at 8,192 shots each. Median $p=1$ raw one-hot feasibility is 65.34%, exact-optimum hit rate is 0.6836%, circuit depth is 43, and two-qubit-gate count is 40. At $p=2$, these become 10.36%, 0.0610%, 448, and 373. The shallow exact-hit median exceeds the 0.3906% uniform-feasible null; the deeper circuit does not.



Hardware depth comparison

All QPU-derived final plans recover the synthetic coordinated gain after exact recourse and validation. Hardware job latency is queue-sensitive: typical medians are about 42-54 seconds, one baseline job waits roughly 11,926 seconds, and another mitigation job reaches about 349 seconds. The corrected end-to-end chart uses a log axis, median labels, and individual-job dots so these outliers remain visible without flattening the other bars.

8. Recommendation, limitations, and submission map

8.1 Deployment recommendation

Use **fast** mode for routine planning, with polished greedy as the normal certified output. Trigger **quality** mode when scarcity, shared resources, high penalty weights, or planner exceptions justify exact LNS latency. Reserve full MILP for bounded comparisons and certificates. Keep **hybrid** mode behind an R&D feature gate with the same exact recourse and validator.

Before production, run a shadow-mode pilot over rolling historical windows. Confirm authorized alternative DCs, working calendars, throughput limits, customer rules, and approval thresholds. Measure plan validity, runtime, planner acceptance, and realized service against default routing.

8.2 Limitations

- The real study uses one supplied planning snapshot; business impact needs rolling and prospective validation.
- The 100-group full-MILP comparator retains a nonzero gap.
- Hardware evidence uses a 16-variable generated control rather than real operations.
- Exact recourse can equalize final outcomes even when proposal quality differs.
- Queue time is external operational latency and can dominate end-to-end hardware time.

8.3 Completed deliverables

Challenge deliverable	Submission artifact
Two-page business/technical summary	reports/business_technical_summary.pdf
Six-to-ten-page technical report	reports/challenge_submission_report.pdf
Runnable repository and notebook	Root README.md; notebooks/nestle_challenge_experiments.ipynb
Five-to-seven-slide presentation	reports/final_presentation.pptx and .pdf
One-page planner view	reports/planner_view.pdf
Mathematical formulation	docs/mathematical_formulation.md
Data and assumptions	docs/data_guide.md; docs/assumptions.md
Scaling, noise, qubit growth, limitations	This report, notebook, and screened figures
Full research treatment	reports/final_report.pdf

The repository also includes unit, integration, notebook, UI, privacy, provenance, and validator regression tests. Raw restricted data and row-level evidence remain local; the branch publishes the implementation and screened aggregate artifacts required to review the approach.